

hw06 - Operator Overloading

Focus

- Operator Overloading

Attachments

- `hw06_start.cpp`. (Of course the file you will turn in will be named `hw06.cpp`.)
- `output.txt`

Task

You will implement a class called **BigUnsigned**. This class will represent unsigned integers that can have **arbitrarily many digits**. In C++, the type unsigned is normally limited to non-negative values less than 2^{32} . You will use a **vector of ints**, with each entry in the vector representing a single digit. (So there is not maximum value that can be held in a BigUnsigned.) Store the digits with the low order digit first, i.e. for the number 123, index 0 of the vector would hold the value 3. This will simplify some of the functions you have to write.

You will provide:

- Constructors:
 - One that can take an int, defaulting to the value 0 if no argument is provided.
 - One that takes a string. The string is intended to be a string of digits.
 - To allow for possible errors, if the string does not start with any digits, then use the value 0.
 - If it starts with a digit, but has other characters, then just use the digits up until the first non-digit character.
 - If there are additional "leading" zeros, they should be skipped or removed.
- Operators
 - Output
 - Addition

- Plus-equals
- Increment operators, both pre- and post-.
- Comparison operators: equals, not-equals, less than, less than or equals, greater than, greater than or equals.
- Follow the conventions that we have discussed to decide whether to implement each operator as a member or non-member function.
- Maximize your code reuse so that it is not necessary to repeat yourself.
- Be sure that the test code that we provide works and will generate the correct output.
 - Maybe we used an operator that wasn't mentioned above?

Be sure to add it!

- Of course provide prototypes for any non-member functions. Naturally if the are declared as friends, you already have the prototype inside of the class.

Do not use:

- string methods such as: `erase`, `first_find_not_of`, `insert`, `stoi`.
- keyword **auto**. Note that you should not use this on any assignment or exam unless the question explicitly instructs you to do so.
- The function **move**. Note that you should not use this on any assignment or exam unless the question explicitly instructs you to do so.

Submit

- **hw06.cpp**

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